1870 → Golgi stained neurons with SILVER

WEEK 2

The Neuron Doctrine → The Nervous System is made of discrete individual cells,

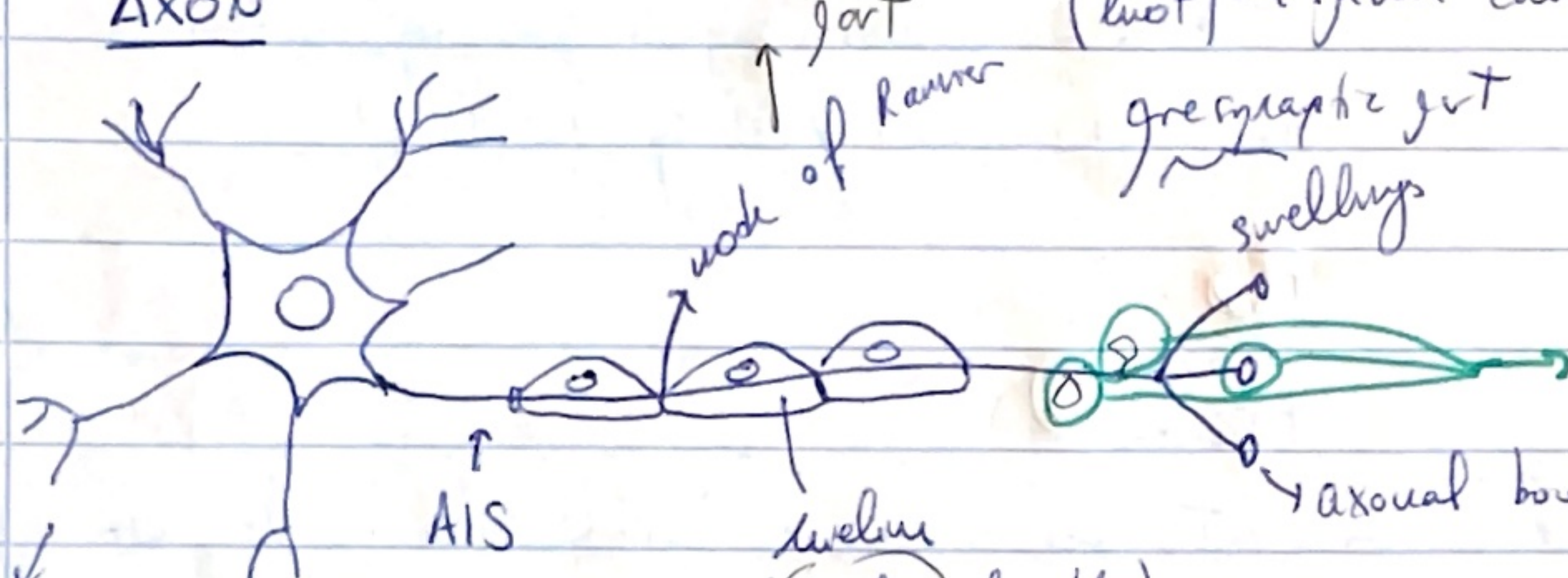
1887 → S. Ramón y Cajal "Neuron Doctrine" → there must be a direction in the flow
"Theory of dynamic polarization"
polarized { Dendrites are the input
Axons are the output

AXON

non isolated part
↑
node of Ranvier

(hot) → spikes can be boosted here

presynaptic part
swellings



Varicosities, where neurotransmitters are (5000/axon) no myelin

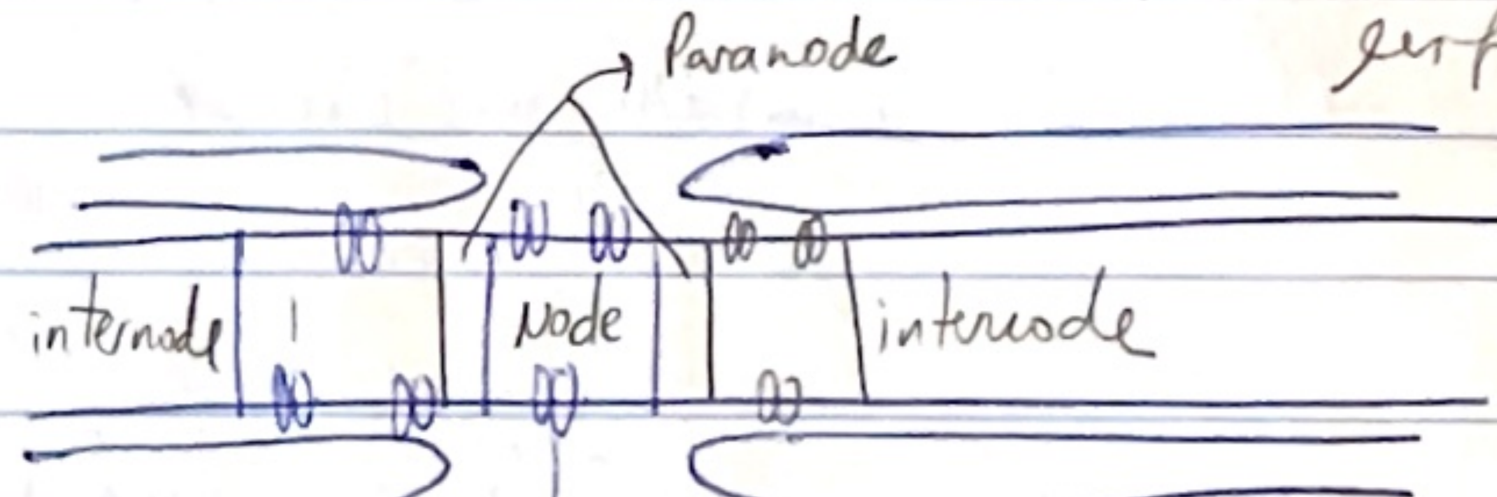
axonal boutons

AIS
Axon initial segment
hot region
generating spikes → where axon potential STARTS
(excitable region) → special ION channels

never have myelin!
↓
if there is myelin
↓
then it's an AXON

myelin
(myelin sheath) → isolating part
created by → oligodendrocytes → lipid
→ Schwann cells in periferal system

Paranode



internode | node | internode

special channels
↓
membrane IONS

[electrical]

DENDRITES

≠ morphology → Purkinje cell (cerebellum) → quite flat
Amacrine cells
Pyramidal cell (hippocampus)

→ spiny dendritic tree

dendritic spines
↓
"appendages" → where synapses are
some neurons are smooth → no spines
10.000/cell

Axon potential
→ Starts in "hot" region

Age of Modern Brain

NEURON TYPES

100 billion of neurons.

- Classification by
  - anatomical feature
  - functional (excitatory vs.
  - electrical / spiking activity
  - chemical characteristics
  - gene expression

do summary last video lesson 2

Principal neurons (+)
from local processing to outer regions (project)

Interneurons (−)
local axonal projection

→ they are also classified.
- morphology
- electrically based

Excitatory → principal projecting cells outside region

Inhibitory → local neurons

Another way of characterising cells is according to how they fire spikes?

SYNAPSE

pre-synaptic axon
post synaptic spine
gap between

$A \xrightarrow{+} D$

Spike allows the transmitter to go out

action potential – axon / (digital – 01, = Bina
post synaptic potential (graded – an

INPUTS
- excitatory → (+) signal
- inhibitory → (−) signal
(interneurons)

↓

all signal go to BODY

Synapse
- strong
- weak
- silent

presynaptic
electric

BMI = Brain Machine Interface

Strength is ratio between these 2 signals

Synapse ⇒ Digital to Analog → weak / strong signal

yes / NO

graded

→ copy slides